Essential Reading: Descriptive Data Sets and Summarizing Data Sets
Reading Objectives:

In this reading, you will be introduced to statistics and descriptive statistics, and various visualizations to understand the data. You will also understand how data that take on only a relatively few distinct values can be described by using frequency tables or graphs or data whose set of values is grouped into different intervals.
Main Reading Section:
Describing data sets
Tables and graphs have been proven to be beneficial ways of presenting data over time, often highlighting essential properties such as the data's range, degree of concentration, and symmetry. This section will look at various typical graphical and tabular data presentation methods.

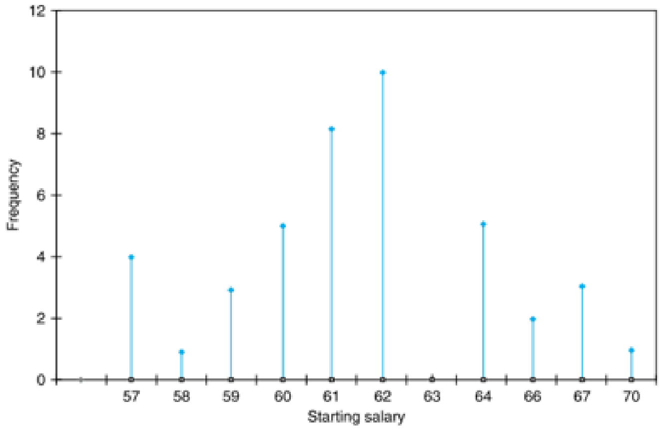
1. Frequency tables and graphs

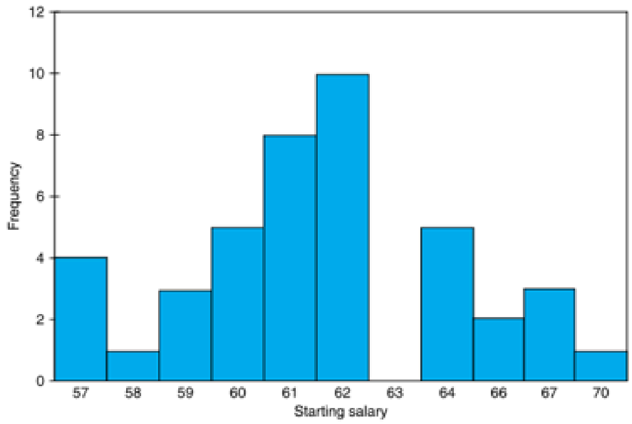
A data set having a relatively small number of distinct values can be conveniently presented in a frequency table.

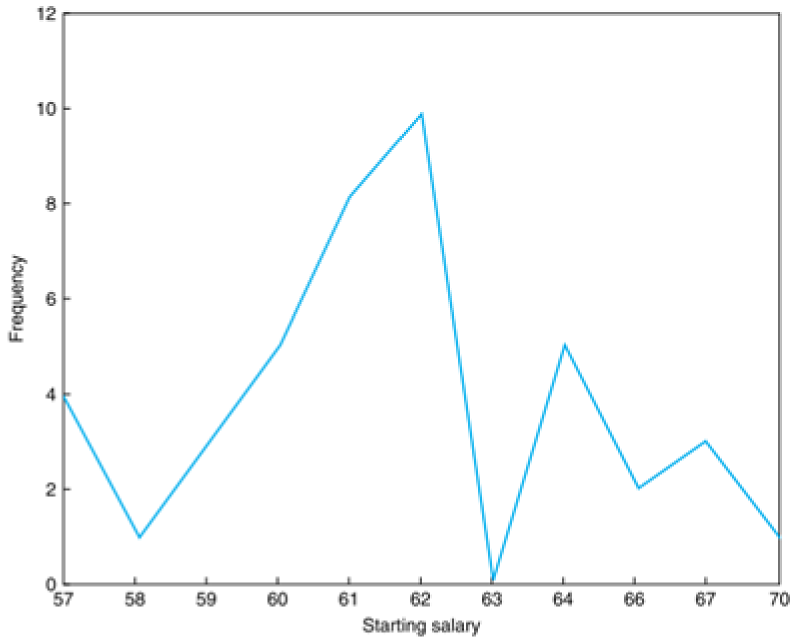
Example 1: Table 1 is a frequency table for a data set consisting of the starting yearly salaries (to the nearest thousand dollars) of 42 recently graduated students with B.S. degrees in electrical engineering. Table 1 tells us, among other things, that the lowest starting salary of $57,000 was received by four of the graduates, whereas the highest salary of $70,000 was received by a single student. The most common starting salary was $62,000 and was received by 10 of the students.

Starting Yearly Salaries	
Starting Salary	Frequency
57	4
58	1
59	3
60	5
61	8
62	10
63	0
64	5
66	2
67	3
70	1

Data from a frequency table can be graphically represented by a line graph, a bar graph, and a frequency polygon, respectively.







2. Histograms, ogives, stem and leaf plots

The number of distinct values in some data sets is too great to use the line or bar graph approach. In such circumstances, it is preferable to divide the values into groups, or class intervals, and then plot the number of data values that fall inside each class interval. The number of class intervals chosen should be a trade-off between:
(i) Selecting too few classes at the expense of losing too much information about the actual data values in a class; and
(ii) Selecting too many classes, which will result in the frequencies of each class being too small to discern a pattern.
The endpoints of a class interval are called the **class boundaries.**

A bar graph plot of class data, with the bars placed adjacent to each other, is called a **histogram.**

The vertical axis of a histogram can represent either the class frequency, which is a **frequency histogram,** or the relative class frequency known as a **relative frequency histogram**.
A cumulative frequency plot is called an **ogive.**

An efficient way of organizing a small- to moderate-sized data set is to utilize a **stem and leaf plot**. Such a plot is obtained by first dividing each data value into two parts—its stem and its leaf.

3. Measures of central tendency

In this section, we introduce some statistics that are used for describing the center of a set of data values.
Suppose that we have a data set consisting of the n numerical values $x_1, x_2, \dots\dots, x_n$. The sample mean is the arithmetic average of these values.
Definition 1: The **sample mean**, designated by $\bar{x}$, is defined by $\bar{x} = \sum_{i=1}^{n} \frac{x_i}{n}$

Note: The computation of the sample mean can often be simplified by noting that if for constants a and b:

$y_i = ax_i + b, \qquad i = 1, 2, \dots\dots., n$

Then, the sample mean of the data set $y_1, y_2, \dots\dots, y_n$ is

$$\bar{y} = \sum_{i=1}^{n} \frac{ax_i + b}{n} = \sum_{i=1}^{n} \frac{ax_i}{n} + \sum_{i=1}^{n} \frac{b}{n} = a\bar{x} + b$$

Example 2. The winning scores in the U.S. Masters golf tournament in the years from 2004 to 2013 were as follows:
280, 278, 272, 276, 281, 279, 276, 281, 289, 280

Find the sample mean of these scores.

Solution: Method 1. Directly adding these values and find the mean.

Method 2. It is easier to first subtract 280 from each one to obtain the new values $y_i = x_i - 280$.
0, −2, −8, −4, 1, −1, −4, 1, 9, 0

The arithmetic average of the transformed data set is $\bar{y} = -\frac{8}{10}$

Which gives $\bar{x} = \bar{y} + 280 = 279.2$

Definition 2: We want to determine the sample mean of a data set that is presented in a frequency table listing the *k* distinct values $v_1, v_2, \ldots \ldots, v_k$, having corresponding frequencies $f_1, f_2, \ldots \ldots, f_k$. Since such a data set consists of $n = \sum_{i=1}^{k} f_i$ observations, with the value v_i appearing f_i times, for each $i = 1, \ldots, k$, it follows that the sample mean of these n data values is $\bar{x} = \sum_{i=1}^{k} \frac{f_i}{n} v_i$.

Definition 3: Order the values of a data set of size n from smallest to largest. If n is odd, the **sample median** is the value in position $(n+1)/2$; if n is even, it is the average of the values in positions $n/2$ and $n/2 + 1$.

Definition 4: The **sample mode** is defined to be the value that occurs with the greatest frequency. If no single value occurs most frequently, then all the values that occur at the highest frequency are called modal values.

Example 3. The following frequency table gives the values obtained in 40 rolls of a die.

Value	Frequency
1	9
2	8
3	5
4	5
5	6
6	7

Find (i) the sample mean, (ii) the sample median, and (iii) the sample mode

Solution:

(i) The sample mean is $\bar{x} = (9 + 16 + 15 + 20 + 30 + 42)/40 = 3.05$.

(ii) The sample median is the average of the 20th and 21st smallest values and is thus equal to 3.

(iii) The sample mode is 1, the value that occurred most frequently.

Reading Summary

In this reading, you have learned the following:

The data set can be represented by a line graph, bar graph, and a frequency polygon.

The data set can be represented by a histogram, ogive, and steam and leaf plots.

The measures of central tendency such as sample mean, median, and mode.